EXPERIMENT -03

```
clc;
clear;
close all;
% Parameters
fs = 100000;              % Sampling frequency
t = 0:1/fs:0.02-1/fs;      % Time vector
fm = 500;                 % Message frequency
fc = 5000;                % Carrier frequency
N = length(t);
% Message Signal
m = cos(2*pi*fm*t);
% Carrier Signal
c = cos(2*pi*fc*t);
% DSB-SC Modulation
dsb = m .* c;
% VSB Filter Design
f = (-N/2:N/2-1)*(fs/N);
H = zeros(1,N);
% Upper Sideband
H(abs(f) >= fc & abs(f) <= fc+fm) = 1;
% Vestigial Lower Sideband
idx = abs(f) >= fc-fm & abs(f) < fc;
H(idx) = (abs(f(idx))-(fc-fm))/fm;
% Apply VSB Filter
DSB_F = fftshift(fft(dsb));
VSB_F = DSB_F .* H;
vsb = real(ifft(ifftshift(VSB_F)));
```

```
% Coherent Demodulation
demod = 2 * vsb .* cos(2*pi*fc*t);
% Low-Pass Filter
[b,a] = butter(4,1000/(fs/2));
rec = filtfilt(b,a,demod);
% Plot Results
figure('Name','VSB Modulation and Demodulation');
subplot(5,1,1);
plot(t,m);
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(5,1,2);
plot(t,c);
title('Carrier Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(5,1,3);
plot(t,dsb);
title('DSB-SC Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
subplot(5,1,4);
plot(t,vsb);
title('VSB Modulated Signal');
xlabel('Time (s)');
```

```
ylabel('Amplitude');
grid on;
subplot(5,1,5);
plot(t,rec);
title('Recovered Message Signal');
xlabel('Time (s)');
ylabel('Amplitude');
grid on;
sgtitle('VSB Modulation and Demodulation');
```

